

ADITYA KULKARNI

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EDUCATION

North Carolina State University

M.S. Computer Science

Raleigh, NC

Aug 2024 – May 2026

University of Mumbai

B.Tech. Computer Science, CGPA: 9.32/10

Mumbai, India

Aug 2020 – May 2024

WORK EXPERIENCE

Product Engineer Intern

May 2025 – Present

Isomer AI

Remote

- Shipped 10+ production backend features for an AI-powered automation platform – Temporal.io workflow orchestration, RBAC, and real-time event pipelines – across 7 deployments using Python, TypeScript, Node.js, PostgreSQL, and Redis
- Built document processing pipelines (PDF generation, Microsoft Graph API, S3) and multi-model LLM tooling (Claude, GPT-4o, Gemini, MCP); contributed 394 commits and led code review across high-velocity Agile cycles

Graduate Research Assistant

Apr 2025 – Present

NC State University

Raleigh, NC

- Engineered a cloud-based genomic data platform on Google Cloud Run with JBrowse 2, enabling interactive visualization of 500GB datasets for 100+ researchers worldwide
- Improved ML model accuracy 12% through hyperparameter tuning and time-series analysis; designed Python-based data pipelines for large-scale biological dataset processing

Software Developer Intern

Jun 2024 – Aug 2024

Mphasis (HP Company)

Remote

- Built 5+ Python-based AI automation workflows including an OCR and NLP pipeline for health claim processing, improving accuracy 25% in production across AWS and GCP deployments

PERSONAL PROJECTS

KubeServe – Kubernetes LLM Inference Platform | Python, FastAPI, Kopf, KEDA, Redis, Helm, ArgoCD

2026

- Built a Python Kopf CRD operator that reconciles LLMDeployment custom resources into Kubernetes Deployments, Services, and KEDA ScaledObjects – enabling declarative, scale-to-zero LLM inference across Claude, GPT-4o, and Gemini backends
- FastAPI router with 3-state circuit breaker and EWMA latency-aware load balancing; OpenTelemetry + Prometheus + Grafana observability; k6 load-tested to 500 VUs; 88 pytest tests; Helm + ArgoCD GitOps

VectorLift – Semantic Search & Ranking Engine | Python, FastAPI, Kafka, Elasticsearch, Qdrant, PostgreSQL

2026

- Built a production-grade semantic search platform: Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant; FastAPI search API with per-stage latency tracking across 6 pipeline configurations
- Fine-tuned bi-encoder (sentence-transformers, in-batch negatives on MS MARCO) and cross-encoder re-ranker (MiniLM); evaluated with NDCG@10, MRR@10, MAP and paired bootstrap significance testing

LLM Eval Harness | Python, Anthropic/OpenAI/Gemini SDKs, DuckDB, Parquet, Streamlit, Docker

2025

- Built a vendor-neutral Python benchmark framework comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 3 task types on a 2,000-example frozen test set with 3 repeated runs per model
- Tracked ExactMatch, TokenF1, hallucination rate, p50/p95 latency, and cost/call; applied paired bootstrap CIs, McNemar test, and Holm correction for statistically rigorous model selection

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, JavaScript, SQL, R, C

Python Ecosystem: FastAPI, PyTorch, TensorFlow, Pandas, NumPy, Pytest, PySpark, LangChain, HuggingFace

Infrastructure: Kubernetes, Docker, KEDA, Helm, ArgoCD, Kafka, Redis, PostgreSQL, AWS, GCP

Observability & Tools: OpenTelemetry, Prometheus, Grafana, GitHub Actions, CI/CD, Terraform